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Unit – 2 : Session – 1 : SLO – 2 | Inserting an Element into an Array

■ I. Pseudocode — Append VAL

- Step 1: Set upper_bound = $N + 1$ (the array grows by one; new upper bound)
- Step 2: Set $A[N + 1] = VAL$ (place the value at the new last position)
- Step 3: EXIT

■ II. Pseudocode — Insert VAL at position POS

- Step 1: [INITIALIZATION] SET $I = N$
- Step 2: Repeat Steps 3 and 4 while $I \geq POS$
- Step 3: SET $A[I + 1] = A[I]$ (shift element right)
- Step 4: SET $I = I - 1$
- [END OF LOOP]
- Step 5: SET $N = N + 1$
- Step 6: SET $A[POS] = VAL$
- Step 7: EXIT
- Note: shifting starts from the last element downward so no element is overwritten before being copied.